

Assignment 1 - Implement TF-IDF and BM25

1. Implement `compute_cosine_similarity` (20 points)
2. Implement `compute_document_tfidf`, analyze its effectiveness (20 points)

ANS 1 & 2:

Question 1 & 2 have been implemented in notebook file

3. The implementation of `compute_document_bm25` is wrong. Fix the bug by **repairing only the most critical error**. Write in your report what's the most critical error. **Do not answer more than 1 errors**; if your answer for Q3 contains multiple errors, you will get 0. (20 points)

ANS 3:

```
tf_d = candidate_word_cnt_dict.get(word, 0)
```

BM25's term frequency must come from the candidate/document, not the query.

4. There is one silly typo in the entire notebook that prevents you from running this notebook til the end, find out where it is, and write it in your report. (20 points).

ANS 4-

Inside the **Running Your Ranking Algorithms** cell the `base_path = pathlib.Path("cs589/assignment1/dataset/")` is the silly typo, this base path has a wrong directory as there is no folder named "cs589" rather the directory is **"cs589assignment1/dataset"**

5. For BM25, by default $k_1=3$, $b=0.75$. Revise the code to tune these parameters based on the MRR, [NDCG@5](#) and [NDCG@10](#) Write in your report with plot, e.g., using matplotlib. Analyze what are the best k_1 and b ? (20 points)

ANS 5:

Best settings (within this grid).

Across all three components on Python, the best pair was **$k_1=2.0$, $b=0.6$**

- **Title:** MRR = 0.4052, NDCG@5 = 0.4218, NDCG@10 = 0.4817 vs default (0.3739, 0.3906, 0.4509). Gains are about **+0.031** absolute on all three metrics ($\approx 7-8\%$ relative), meaning relevant duplicates appear earlier in the ranked list and the top-k quality improves.

- **Question:** 0.3043 / 0.3157 / 0.3728
vs default 0.3043 / 0.3157 / 0.3728. **No meaningful change.**
- **Answer:** 0.2661 / 0.2695 / 0.3296
vs default 0.2661 / 0.2695 / 0.3296. **No meaningful change.**

title best mrr: k1=2.0, b=0.6, score=0.4052

title best ndcg5: k1=2.0, b=0.6, score=0.4218

title best ndcg10: k1=2.0, b=0.6, score=0.4817

question best mrr: k1=2.0, b=0.6, score=0.3043

question best ndcg5: k1=2.0, b=0.6, score=0.3157

question best ndcg10: k1=2.0, b=0.6, score=0.3728

answer best mrr: k1=2.0, b=0.6, score=0.2661

answer best ndcg5: k1=2.0, b=0.6, score=0.2695

answer best ndcg10: k1=2.0, b=0.6, score=0.3296